

figures/logo.png

EI-SCORE

Explainable-Interpretable and Simple
Customized Overall Ranking Engine

User Guide

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*An explainable and interpretable composite indicator based on decision
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Introduction

EI-SCORE is an interactive, browser-based application for building explainable and interpretable composite indicators using decision rules. It implements the Dominance-based Rough Set Approach (DRSA) as described in Corrente et al. (2026).

The tool is designed for researchers, analysts, and decision-makers who need to rank or classify a set of alternatives (countries, companies, students, patients, etc.) evaluated on multiple criteria, and who require transparent, human-readable justifications for the rankings produced.

Key features

- **Classification problems** — assigning alternatives to ordered classes (e.g., low, medium, high)
- **Scoring problems** — ranking alternatives by a continuous score
- **Rule induction** — inducing at-least ($\mathcal{R}$) and at-most ($\mathcal{R}$) decision rules from reference units
- **Full pipeline** — running the complete 7-step method including greedy selection and MILP minimisation
- **New unit classification** — classifying new alternatives via MILP Problems (6), (7), and (8)
- **Rule reuse** — applying saved rules to new datasets without re-running the pipeline
- **Missing value support** — automatic detection and handling via Algorithm 4

Reference

Corrente S., Greco S., Słowiński R., Zappalà S. (2026). *An explainable and interpretable composite indicator based on decision rules*. Omega, 142, 103513.
<https://doi.org/10.1016/j.omega.2026.103513>

Installation and Launch

EI-SCORE can be used in two ways: as a locally installed application or via a web browser without any installation.

Option A — Web access (no installation required)

The application is publicly available online. Simply open the following URL in your browser:

<https://github.com/silvanozap/Explainable-composite-indicator>

No Python or package installation is required.

Option B — Local installation

Requirements:

- Python 3.14.3 or later
- numpy==2.4.2, pandas==3.0.1, scipy==1.17.0, streamlit==1.58.0

Step 1 — Clone the repository:

```
git clone https://github.com/silvanozap/Explainable-composite-indicator.git
cd Explainable-composite-indicator
```

Step 2 — Install dependencies:

```
pip install -r requirements.txt
```

Step 3 — Launch the application:

```
streamlit run drsa/interface/app.py
```

The application opens automatically at <http://localhost:8501>.

Input File Formats

Data file

The data file must be a CSV or TXT file with the following structure:

```
Name,criterion_1,criterion_2,...,criterion_k,class
A1,4,3,2,1
A2,5,4,3,2
A3,7,6,5,
A4,6,5,4,
```

- **First column** (optional): unit names. If the first column contains non-numeric values, it is automatically interpreted as unit names.
- **Criteria columns**: numeric values. Any column separators (comma, semicolon, tab, space) are supported and selectable in the interface.
- **Last column**: class label.
 - Integer value → **reference unit** (used for rule induction)
 - Empty / NaN → **non-reference unit** (classified but not used for induction)
- **Missing values**: empty cells in criteria columns are detected automatically. When present, Algorithm 4 is applied instead of Algorithm 1/2.

Tip

Sample data files covering all use cases are available in the **examples/** folder of the repository, and can also be downloaded directly from the application sidebar.

Rules file (classification)

Rules exported from the application or created manually follow this format:

```
#directions,increasing,decreasing,...
#mode,class
type,class,criterion_1,criterion_2,...
at-least,2,0.762,,
at-most,1,,2.71,
```

- **#directions:** preference direction of each criterion (**increasing** or **decreasing**)
- **#mode:** **class** for classification problems
- **Data rows:** each row is a rule with **type** (**at-least** or **at-most**), **class** (assigned class), and threshold values for each criterion (empty if the criterion is not in the rule condition)

Rules file (scoring)

For scoring problems, the **#mode** line changes and class values represent score values:

```
#directions,increasing,decreasing,...
#mode,score
type,class,criterion_1,criterion_2,...
at-least,16.67,0.762,,
at-most,0,,2.71,
```

Tip

Sample rules files for both classification and scoring can be downloaded directly from Tab 5 (Apply Rules) of the application.

Tab 1 — Data & Setup

The Data & Setup tab is the starting point of every analysis. It allows the user to load data, configure the problem, and set all analysis parameters.

[Screenshot: Tab 1 — Data & Setup overview]

Figure 1: Tab 1 — Data & Setup

Upload data

Click the **Upload CSV or TXT** button in the sidebar to load your data file. Select the appropriate column separator (comma, semicolon, tab, or space) from the dropdown. Once uploaded, the dataset is displayed in the main area.

Problem type

Select the problem type using the **Problem type** radio button:

- **Classification** — the last column contains integer class labels
- **Scoring** — the last column contains continuous score values. The tool automatically maps scores to ordinal class indices internally; all rule texts and classification results are displayed using the original score values.

Reference units

Reference units are the alternatives whose class assignments are known and are used for rule induction. By default, all units with a non-empty class label are considered reference units. You can manually select a subset of reference units using the multiselect widget.

Tip

Non-reference units (empty class) are classified by the induced rules but do not participate in rule induction.

Preference directions

For each criterion, select the preference direction:

- **Increasing** ($\uparrow$) — higher values are preferred (e.g., profit, life expectancy)
- **Decreasing** ($\downarrow$) — lower values are preferred (e.g., cost, risk)

[Screenshot: preference direction selection]

Figure 2: Preference direction selection

Settings

- **Min confidence (c)**: minimum confidence threshold for rule induction (default: 1.0). Reducing this value allows the induction of approximate rules.
- **Random seed**: controls the random number generator used in the greedy selection step (Steps 3–4) for reproducibility.

Tab 2 — Run

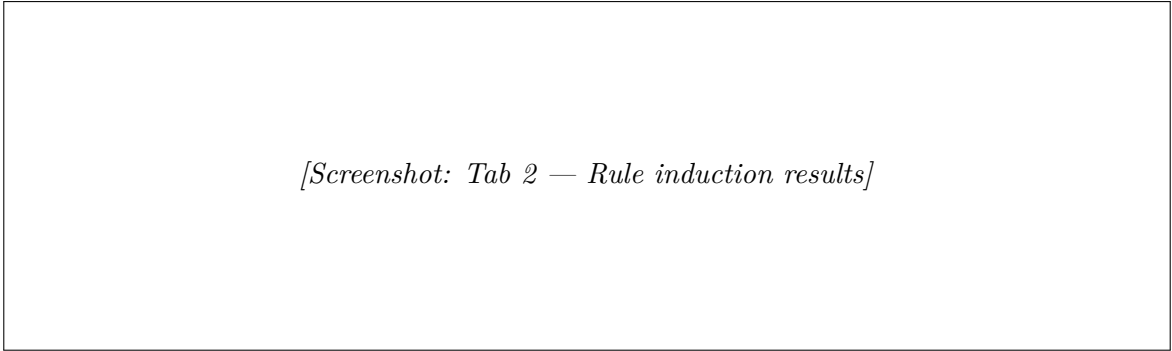
The Run tab is where rule induction and the full pipeline are executed.

Rule induction only

Select **Rule induction only** to run Steps 1–2 of the algorithm: induction of at-least ($\mathcal{R}$) and at-most ($\mathcal{R}$) decision rules from the reference units using Algorithms 1, 2, or 4.

The induced rules are displayed in collapsible expanders, showing:

- Rule text with condition and class assignment
- Support value (fraction of reference units covered)
- Matching units (units satisfying the rule condition)



[Screenshot: Tab 2 — Rule induction results]

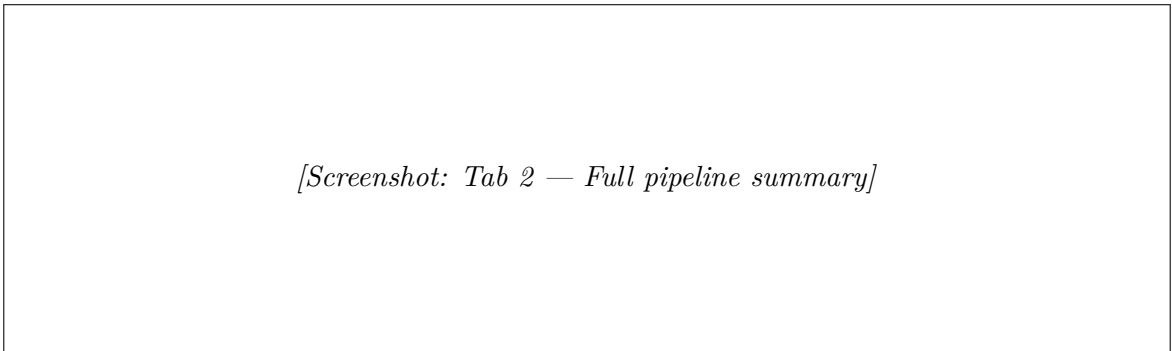
Figure 3: Tab 2 — Rule induction results

Full pipeline

Select **Full pipeline** to run all 7 steps:

1. **Steps 1–2:** Rule induction from reference units
2. **Steps 3–4:** Greedy selection of rules that avoid contradictory classification of all units
3. **Step 5:** Classification of all units using equation (4)
4. **Step 6:** Rule induction from all units (maximal rule set)
5. **Step 7:** MILP-based minimisation (minimal rule set)

A progress bar shows the pipeline execution. Upon completion, a summary table shows the number of rules at each step.



[Screenshot: Tab 2 — Full pipeline summary]

Figure 4: Tab 2 — Full pipeline summary

Exporting rules

After running, the following CSV files can be downloaded:

- **Minimal rules CSV:** the MILP-minimised rule set (Step 7)
- **Maximal rules CSV:** the full rule set from all units (Step 6)

Tab 3 — Classification

The Classification tab displays the classification of all units using equation (4) from the paper.

Classification table

For each unit, the table shows:

- s^- : lower class bound (best class supported by at-least rules)
- s^+ : upper class bound (worst class supported by at-most rules)
- **Assignment**: classification result
 - $s^- = s^+$: unit assigned to a single class
 - $s^- < s^+$: unit assigned to a range of classes (uncertain)
 - $s^- > s^+$: contradictory classification
- **Status**: OK or Contradictory

[Screenshot: Tab 3 — Classification table]

Figure 5: Tab 3 — Classification table

Unit-by-unit explanation

Select a unit from the dropdown to see a detailed explanation of its classification: which rules were matched, and how s^- and s^+ were determined.

[Screenshot: Tab 3 — Unit-by-unit explanation]

Figure 6: Tab 3 — Unit-by-unit explanation

Tab 4 — New Units

The New Units tab allows classifying new alternatives that were not part of the original dataset, using MILP Problems (6), (7), and (8) to handle contradictions.

Uploading new units

Upload a CSV or TXT file containing the new alternatives. The file must include the same criteria columns as the original dataset, **without** a class column. Unit names in the first column are optional.

Warning

The Full pipeline must be run (Tab 2) before new units can be classified.

Classification results

The results table shows all original units together with the new units (highlighted in yellow), displaying:

- $s^-(prev)$: initial classification with maximal rules (before MILP)
- $s^+(prev)$: initial upper bound
- $s^-(new)$: final classification after MILP resolution
- $s^+(new)$: final upper bound
- **Assignment**: final class assignment
- **Changed**: whether the classification changed from the initial assignment

[Screenshot: Tab 4 — New units classification table]

Figure 7: Tab 4 — New units classification results

Unit-by-unit explanation

The unit-by-unit explanation is available for all units (original and new), using the minimal rule set found after MILP Problem (8).

Tab 5 — Apply Rules

The Apply Rules tab allows loading a previously saved rules file and applying it to new alternatives, without needing to re-run the pipeline.

Tip

This tab is always accessible, even without loading a data file in the sidebar.

Loading rules

Upload a rules CSV file (in the format described in Section 3). The application automatically reads the `#directions` and `#mode` lines to reconstruct the criterion directions and problem type.

Upon loading, the application displays:

- Number of at-least and at-most rules
- Criteria names and directions

Visualising rules

The induced rules are displayed with the matching units highlighted (if alternatives have been loaded).

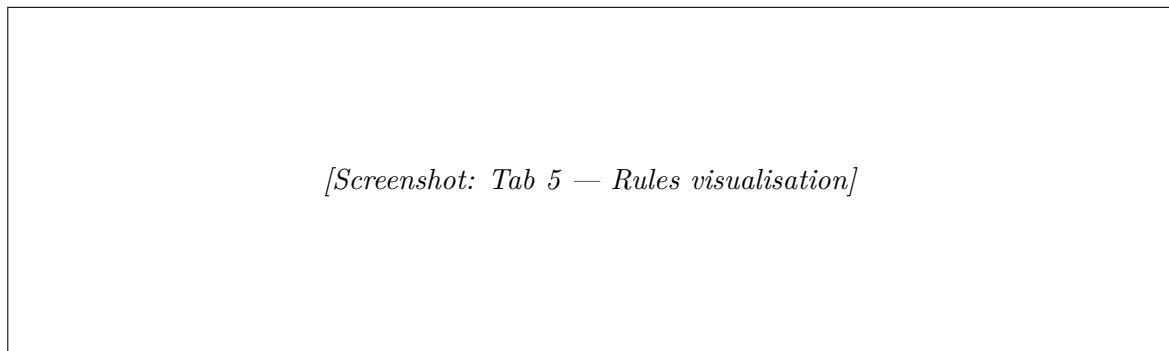


Figure 8: Tab 5 — Rules visualisation

Loading alternatives and classification

Optionally upload a file of alternatives (without class column) to classify them using the loaded rules. The classification table and unit-by-unit explanation are displayed as in Tab 3.

Warning

The column names in the alternatives file must exactly match the criterion names in the rules file.

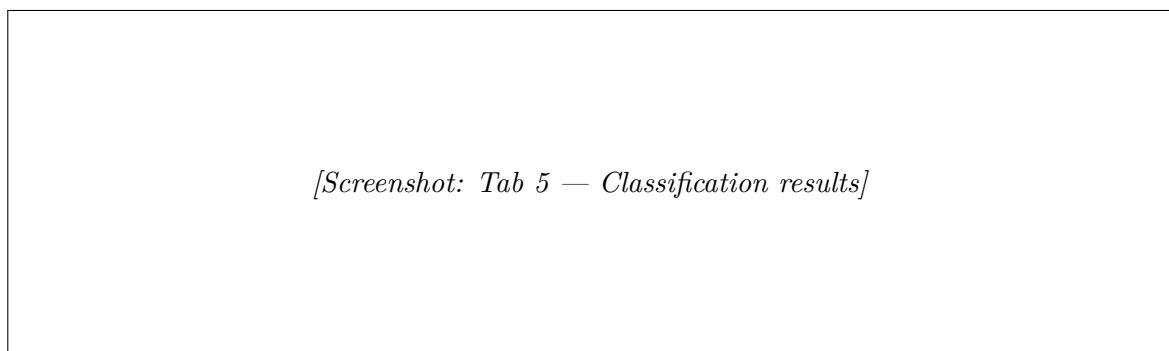


Figure 9: Tab 5 — Classification results

Examples

All example files are available in the `examples/` folder of the repository.

Example 1 — Human Development Index (HDI)

Files: `examples/HDI.csv`

Problem: Classification

Dataset: 193 countries evaluated on 4 criteria (life expectancy, expected years of schooling, mean years of schooling, GNI per capita), assigned to 4 development classes.

Directions: all increasing ($\uparrow$)

Workflow:

1. Upload `HDI.csv` in the sidebar
2. Tab 1: set all criteria as increasing
3. Tab 2: select *Rule induction only* and click Run
4. Tab 2: inspect the induced rules
5. Tab 3: inspect the classification of all countries

Example 2 — Stock Portfolio (PTF)

Files: `examples/PTF.csv`, `examples/PTF_new_units.csv`

Problem: Full pipeline + New units

Dataset: 50 stocks evaluated on 8 financial criteria (Return, Beta, Net Profit Margin, ROA, ROE, EPS, P/E, P/BV), assigned to 3 performance classes.

Directions: all increasing ($\uparrow$)

Workflow:

1. Upload `PTF.csv` in the sidebar
2. Tab 1: set all criteria as increasing
3. Tab 2: select *Full pipeline* and click Run
4. Tab 3: inspect classification of all stocks
5. Tab 4: upload `PTF_new_units.csv` and classify new stocks

Example 3 — ElectreScore (Scoring problem)

Files: `examples/ElectreScore.csv`

Problem: Scoring

Dataset: 10 investment projects evaluated on 5 criteria, ranked by a continuous score from 0 to 100.

Directions: $g1 \downarrow$, $g2 \downarrow$, $g3 \uparrow$, $g4 \uparrow$, $g5 \uparrow$

Workflow:

1. Upload `ElectreScore.csv` in the sidebar
2. Tab 1: select *Scoring* as problem type; set criterion directions
3. Tab 2: select *Rule induction only* and click Run
4. Tab 2: inspect rules — note that class labels show original score values
5. Tab 3: inspect classification — s^- and s^+ show score values

Example 4 — Missing Values (PISA dataset)

Files: `examples/MISSING.csv`, `examples/MISSING_new_units.csv`

Problem: Classification with missing values, Full pipeline

Dataset: 20 students evaluated on 3 criteria (mathematics score, reading score, socioeconomic index), assigned to 3 performance classes. Several missing values are present.

Directions: all increasing ($\uparrow$)

Workflow:

1. Upload `MISSING.csv` in the sidebar
2. Tab 1: set all criteria as increasing — Algorithm 4 is applied automatically
3. Tab 2: select *Full pipeline* and click Run
4. Tab 3: inspect classification
5. Tab 4: upload `MISSING_new_units.csv` and classify new students

Example 5 — Apply Rules (rule reuse)

Files: `examples/rules_score.csv`, `examples/ElectreScore.csv`

Problem: Rule reuse, scoring

Dataset: Pre-computed rules for the ElectreScore dataset (26 at-least, 25 at-most rules, 5 criteria, score mode).

Workflow:

1. Go directly to Tab 5 (no data upload needed)
2. Upload `rules_score.csv`
3. Inspect the rules with criterion directions and score labels
4. Upload `ElectreScore.csv` as alternatives
5. Inspect classification — s^- and s^+ show original score values

Troubleshooting

File format errors

- **“Could not read file”:** the file separator does not match the selected separator. Try changing the separator dropdown (comma, semicolon, tab, space).
- **Criteria columns show unexpected values:** ensure that the class label is in the **last** column of the file. All intermediate columns are treated as criteria.
- **Non-numeric values in criteria:** columns with text values are treated as unit names (first column) or ignored. Ensure all criteria columns contain numeric values only.
- **No reference units detected:** if all class labels are empty, no rules can be induced. Ensure that at least some units have a numeric class label in the last column.

Rules file errors

- **“Missing #directions line”:** the rules file must start with a `#directions` line. Check the file format in Section 3.

- **Column names do not match** (Tab 5): when classifying alternatives in Tab 5, the column names in the alternatives file must exactly match the criterion names in the rules file (case-sensitive, no extra spaces).
- **Wrong #mode:** if rules were exported from a scoring problem, the file contains `#mode,score`. Loading it in Tab 5 will display score values instead of class indices. This is correct behaviour.

Pipeline errors

- **“Run the Full pipeline first”** (Tab 4): the New Units tab requires the full pipeline to be completed first. Go to Tab 2, select *Full pipeline*, and click Run.
- **MILP takes a long time:** for large datasets with many contradictions, the MILP optimisation (Step 7 and Problems 6–8) may take several minutes. This is expected behaviour.

Acknowledgements
